

The Propose Currency Detection Technique Based on Sift Algorithm and Bayesian Classifier: A Review

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Abstract

In this topic we are working to check the currency of India means we will apply the technique feature matching and recognize whether that currency notes are genuine or not. In the previous research SIFT algorithm is applied for feature matching. This algorithm only matches colour and paper type but do not match internal features like data which is printed on the strip. In future, techniques will be proposed which match the internal features of the note and recognize that whether it is genuine or not. The proposed enhancement will lead to increase efficiency of reorganization. The main advantage of this research is that it leads to design of automation system to check that currency is genuine or not.

Keywords: SIFT algorithm, currency detection, image processing

I. INTRODUCTION

The paper currency constitutes extrinsic as well as intrinsic properties. The extrinsic properties deals with size, width, colour, etc. whereas intrinsic properties deals with security thread, I.D. mark, number panel, etc. Extrinsic properties are not enough to recognize whether note is original or fake. Also currency may get damaged during transportation or exchange. The change in nature of an image can be well understood and improved with the help of image processing techniques. It enhances pictorial information embedded in the image for human interpretation [1]. RBI is the only body which has sole responsibility to print currency notes in India. Every year RBI faces the

Problem of counterfeit currency notes once filtered and circulated in the market. Earlier fake currency detection was done by using chemical Properties of the currency paper. With the advent of artificial intelligence and image processing, digital image processing is frequently used for fake currency detection by extracting attributes from images that represent the features of currency [2].

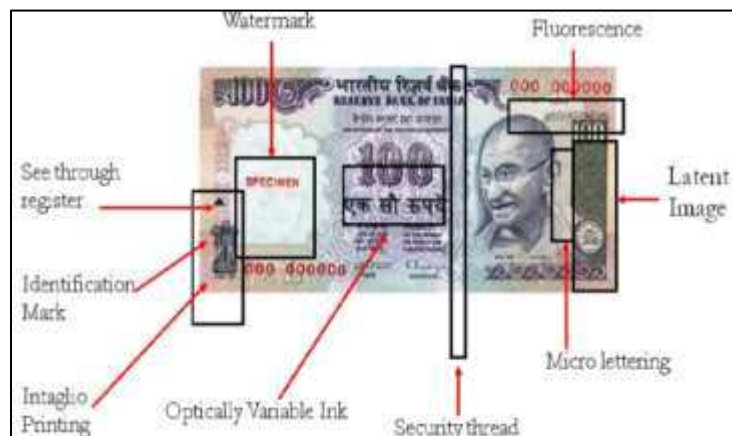


Fig. 1: Indian currency notes [6]

Automated paper currency recognition system can be a very good utility in banking systems and other field of commerce. Since many years counterfeiting of paper currency challenges the financial system of every country in different sectors, India is also one of them. Modernization of the financial system is a milestone in protecting the economic prosperity, and maintaining social harmony. Automatic machines capable of recognizing banknotes are massively used in automatic dispensers of a number of different products, ranging from cigarettes to bus tickets, as well as in many automatic banking operations. The needs for automatic banknote recognition systems encouraged many researchers to develop corresponding robust and reliable techniques. The technology of currency recognition aims to search and extract the visible and hidden marks on paper currency for efficient

classification. Until now, there are many methods proposed for paper currency recognition. The simplest way is to make use of the visible features of the paper currency, for example, the size and color of the paper currency. However, this kind of methods has great limitations as banknotes are getting worn and torn with the passing of time and they are even dirtier when holding by dirty hands or in dirt. If any banknote is dirty or it may be changed into any other color then the color content of banknote may change largely [3]. Security features of Indian currency notes [6]

A. See through Register

The small floral design printed both on the front (hollow) and back (filled up) of the note in the middle of the vertical band next to the Watermark has an accurate back to back registration. The design will appear as floral design when seen against the light [6].

B. Water Marking

The Mahatma Gandhi Series of banknotes contain the Mahatma Gandhi watermark with a light and shade effect and multi-directional lines in the watermark window [6].

C. Optically Variable Ink

This is a new feature included in the Rs.1000 and Rs.500 notes with revised color scheme introduced in November 2000. The numeral 1000 and 500 on the obverse of Rs.1000 and Rs.500 notes respectively is printed in optically variable ink viz., a color-shifting ink. The colour of the numeral 1000/500 appears green when the note is held flat but would change to blue when the note is held at an angle [6].

D. Fluorescence

Number panels of the notes are printed in fluorescent ink. The notes also have optical fibers. Both can be seen when the notes are exposed to ultra-violet lamp [6].

E. Security thread

The Rs.500 and Rs.100 notes have a security thread with similar visible features and inscription „Bharat“ (in Hindi), and „RBI“. When held against the light, the security thread on Rs.1000, Rs.500 and Rs.100 can be seen as one continuous line. The Rs.5, Rs.10, Rs.20 and Rs.50 notes contain a readable, fully embedded windowed security thread with the inscription „Bharat“ (in Hindi), and „RBI“. The security thread appears to the left of the Mahatma's portrait [6].

F. Intaglio printing

The portrait of Mahatma Gandhi, the Reserve Bank seal, guarantee and promise clause, Ashoka Pillar Emblem on the left, RBI Governor's signature are printed in intaglio i.e. in raised prints, which can be felt by touch, in Rs.20, Rs.50, Rs.100, Rs.500 and Rs.1000 notes [6].

G. Latent Image

On the obverse side of Rs.1000, Rs.500, Rs.100, Rs.50 and Rs.20 notes, a vertical band on the right side of the Mahatma Gandhi's portrait contains a latent image showing the respective denominational value in numeral. The latent image is visible only when the note is held horizontally at eye level [6].

H. Micro Lettering

This feature appears between the vertical band and Mahatma Gandhi portrait. It always contains the word „RBI“ in Rs.5 and Rs.10. The notes of Rs.20 and above also contain the denominational value of the notes in micro letters. This feature can be seen well under a magnifying glass [6].

I. Identification Mark

Each note has a unique mark of it. A special feature in intaglio has been introduced on the left of the watermark window. This feature is in different shapes for various denominations (100-Triangle, Rs.500-Circle, and Rs.1000- Diamond) and helps the visually impaired to identify the denomination [6].

Digital Image Processing is a field that involves studying as well as processing of images by extracting attributes from images and includes the recognition of individual objects. Extracting attributes for images of currency notes can get quite complex as it involves the extraction of some visible and invisible features of Indian currency [2].

II. RELATED WORK

They defined that currency reorganization is one of the fast growing research fields under image processing. This paper introduces a method for currency reorganization. The main objective of this scheme to identify features like Numeral, Shape, colors etc.

Principal Component Analysis is used to reduce the dimensions and a similarity based classifier is constructed to predict test sample. Results are also validated by constructing models using classifier implemented using WEKA and testing with unseen samples not considered in feature extraction. This study demonstrated that center numeral results in an accuracy of 100% with all family of currencies [4].

In this paper they examined counterfeit notes are the one of the biggest problem in cash transactions. It has become one of the biggest hurdles for INDIA. Due to easy printing, it is very easy to print fake news within seconds using latest tools. Detecting fakes notes with manually process is time consuming and vary untidy. So there is a need of a machine that automatically detects fake currency in efficient manners. Many techniques have been proposed with the use of MATLAB, feature extraction with HSV color space and other applications of image processing. We have implemented a fake note detection unit with MATLAB algorithm. This paper is a based on the same project to give solution for fake currency problem [5]. Even though denomination based on size may or may not be identified but it is almost difficult to identify whether the note is original or fake. It is the question on edge to develop such a system that will make sure for visually disabled or blind people that the currency they have is original or not. The currency recognition algorithm discussed in this paper using image processing is based on an ORB (Oriented fast and Rotated Brief). It is faster and also rotation invariant. The proposed algorithm shows 90% true recognition rate [1].

Main aim of this paper is to detect fake currency using image processing. Fake currency detection is a process of finding the forgery currency. After choose the image apply preprocessing. In pre-processing the image to be crop, smooth and adjust. Convert the image into gray color. After conversion apply the image segmentation. The features are extracting and reduce. Finally compare the image into original or forgery [6].

They are designing a system that helps in identification of Indian currency notes and to check whether it is a valid or invalid. This is to differentiate between the counterfeit notes and genuine notes. Currency features such as See Through register, See Through register symbol or Identification mark, Security thread, Governor's signature, Micro lettering, year of print. These features are segmented using 3x3 grids. This is done by the use of SIFT technique which helps in efficient matching of the features [3].

III. CONCLUSION

In this topic we are working to check the currency of India means we will apply the technique feature matching and recognize weather that currency notes are genuine or not. In the previous research SIFT algorithm is applied for feature matching. This algorithm only matches colour and paper type but do not match internal features like data which is printed on the strip. In future, techniques will be proposed which match the internal features of the note and recognize that whether it is genuine or not. The proposed enhancement will lead to increase efficiency of reorganization. The main advantage of this research is that it leads to design of automation system to check that currency is genuine or not

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